Paper 3: Ledgers

Admissible Dynamics, Entropy, and the Arrow of Time

v3.0

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Admissibility Physics — March 2026 (v3.0 scope expansion: April 2026)

Abstract

Paper 2: *The Structure of Admissible Physics* established that the correlation space has a unique architecture: a gauge group SU(3) × SU(2) × U(1), a fermion content of 45 Weyl fermions in three generations, and a capacity partition $42 + 19 = 61$ that fixes the cosmological density fractions. The space has rooms, and we can count them. What it does not yet have is a direction.

This paper supplies the arrow.

We derive three results from the foundation of Paper 1, all following from the four structurally necessary components of the *Principle of Least Enforcement Cost* (PLEC) – A1 (finite enforcement capacity, upper bound), MD (positive cost floor), A2 (argmin selection), BW (non-degeneracy) – together with the tensor product structure established in Paper 1 (T_{tensor}):

1. **Complete positivity as a theorem (T_{CPTP}):** Admissibility-preserving evolution maps are exactly the completely positive, trace-preserving (CPTP) maps. This closes the conditional result deferred from Paper 1, where ancilla-stability required analysis of how admissible maps compose under tensor products. The Stinespring dilation [5] receives a direct enforcement interpretation: system and environment evolve unitarily with separate budgets, then the environment budget is traced out.
2. **Entropy as committed capacity (T_{entropy}, T_κ):** The von Neumann entropy of a state equals the enforcement demand of its active correlations at the system interface. The directed enforcement multiplier $\kappa = 2$ is derived (not postulated) from irreversibility and locality. The zeroth and first laws of thermodynamics follow as corollaries.
3. **The arrow of time from irreversibility (L_{irr}, $T_{\text{second law}}$):** Capacity committed to system-environment correlations cannot be recovered by any local operation. This is not a boundary condition; it is forced by A1, L_{nc}, and L_{loc}. The direction of non-decreasing entropy is the arrow of time. T-violation of phase $\pi/4$ quantifies the asymmetry.

These results sit inside the variational framework that Papers 1 and 2 now present as the *Principle of Least Enforcement Cost* (PLEC): reality as the minimum-cost expression of distinction compatible with finite capacity, with A1 (finite capacity; upper bound), MD (positive cost floor), A2 (argmin selection), and BW (non-degeneracy) as its four structurally necessary components. Paper 3 introduces no new components; the results above are downstream consequences of the same four.

A secondary result, derived from the capacity-counting structure of Paper 2, identifies the Fisher information metric on the space of admissible states as a direct consequence of the entropy functional, and shows that RG flow is Fisher gradient descent. Two distinct CP-violation mechanisms emerge from the single irreversibility lemma L_{irr}: holonomy phases (CKM) and entropy-optimization phases (PMNS).

Paper 3 also serves as the entropy paper at cosmological scale. Enforcement cost is the quantum-interface face of a more general capacity concept—the *admissibility carrying capacity* (ACC) of a relational interface, whether that interface is a bipartite quantum split,

a causal horizon, or the cosmological boundary. Part II closes with a geometric extension that recovers horizon entropy (prefactor 1/4) as a special case of the same ledger accounting; Part III extends the arrow of time to a macroscopic regime where the arrow is identified with loss of global admissible coordination, and closes with a structural treatment of cosmogenesis in which low initial entropy is reframed as restricted admissible phase space and cosmogenesis proceeds by staged constraint relaxation.

No new axioms are introduced. All results trace to named check functions in the codebase. The paper is structured in three parts: Part I (admissible dynamics), Part II (entropy, thermodynamics, and geometric carrying capacity), and Part III (time, cosmogenesis, information geometry, and CP structure).

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1. Introduction: the correlation space needs a direction

Paper 1 and Paper 2 together built a static picture. Paper 1 showed that finite enforcement capacity forces a quantum operator algebra, a state space, and a gauge symmetry. Paper 2 showed that the only algebra compatible with A1 is $\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)$ acting on the Standard Model fermion content, and that the capacity budget partitions into 42 vacuum channels and 19 matter channels. The correlation space $\mathcal{S}_{\mathrm{adm}}$ has been mapped, measured, and classified.

But a map is not a movie. The space as described so far has no preferred direction, no thermodynamic interpretation, and no intrinsic notion of past versus future. Time appears as an external parameter. Entropy is a functional on states, but nothing has been said about how it changes, or why it changes in a particular direction.

This paper shows that both the dynamics and the arrow arise from the same source as everything else: the four-component variational structure that Papers 1 and 2 call the *Principle of Least Enforcement Cost* (PLEC), of which A1 is the upper-bound component.

That source can be stated compactly. PLEC – reality as the minimum-cost expression of distinction compatible with finite capacity – packages A1 (upper bound), MD (positive floor), A2 (argmin), and BW (non-degeneracy) as four structurally necessary components of a single variational principle. PLEC is the canonical variational formulation of the finite-enforceability framework; it does not replace the locked claim that physical meaning is grounded in enforceable distinction. In this paper, "A1" continues to denote the finite-capacity component; the argmin, floor, and non-degeneracy components (A2, MD, BW) enter where selection and separation are at stake.

The argument has a natural structure. Before we can ask how the correlation space *evolves*, we need to characterize which evolution maps are admissibility-preserving. That is Section 4: the answer is the CPTP maps, as a theorem rather than a definition. Once we know what evolution looks like, we can ask what it does to entropy: Section 6 shows that entropy equals committed capacity, and derives the directed-enforcement multiplier $\kappa = 2$ from locality and irreversibility. The zeroth and first laws of thermodynamics follow.

The second law requires a different argument, because it is not a statement about what maps *preserve* but about what the world *does*. Section 7 derives it from L_{irr}: the lemma that cross-interface correlations commit capacity that no local observer can recover. Section 10 shows that this lemma selects an orientation on the single timelike direction forced by the Lorentzian signature of spacetime (Paper 5), and that T-violation at phase $\pi/4$ (T_{CPT}) quantifies the asymmetry.

Sections 12 and 13 are consequences rather than new derivations. Section 12 shows that the entropy functional carries a natural Riemannian metric—the Fisher information metric—and that RG flow is gradient descent on the information geometry. Section 13 shows that L_{irr} generates two structurally distinct CP-violation mechanisms that correspond exactly to the CKM and PMNS phases derived in Paper 4.

Between the quantum-interface entropy results of §§ 4–6 and the information-geometry arguments of §§ 10–11, two extensions broaden the scope. Section 8 extends the ledger formalism to geometric interfaces: the admissibility carrying capacity (ACC) of a causal boundary obeys the same accounting as the enforcement cost of a quantum bipartition, and horizon entropy $S_H = A/4$ emerges from two structurally forced bookkeeping factors. Section 11 applies the resulting framework at cosmological scale: the constraint functional $\Phi_c = \ln(d_{\mathrm{field}}/d_{\mathrm{admissible}})$

quantifies phase-space suppression, the earliest admissible regime is overconstrained, and cosmogenesis proceeds by staged release of degrees of freedom. Neither section introduces a new axiom; both are consequences of A1 applied to finite interfaces at different scales.

1.1 What this paper does not claim

Several important results are previewed but not fully derived here.

- **The full CPTP derivation at composite interfaces.** T_{CPTP} is proved here for single-system evolution. The extension to composite systems at shared interfaces requires the locality structure developed in Paper 5. The present proof is complete and unconditional for the setting it addresses.
- **Quantitative CP phases (CKM and PMNS).** Section 13 identifies the mechanism that generates the two phase classes; the numerical derivation of δ_{CKM} and δ_{PMNS} is carried out in Paper 4, which has access to the generation structure and mass hierarchy.
- **The cosmological constant and expansion history.** $T_{\text{second law}}$ establishes that entropy saturates at $S_{dS} = 61 \ln 102$ nats and §8 identifies this saturation with the de Sitter horizon's ACC saturation. The quantitative relation $\Lambda \cdot G = 3\pi/102^{61}$, the confrontation with DESI DR2, and the dynamical-dark-energy tension are the subject of Paper 6.
- **The confinement and mass scales.** The entropy and Fisher results in §§ 12–13 establish structural connections to the mass hierarchy; quantitative predictions require Paper 4's treatment of the Frobenius-Nagata chain.

1.2 Derivation chain for this paper

The derivation chain from A1 to the main results is shown in Figure 2. All steps use results proved in Paper 1 or Paper 2; no new axioms are required at any stage.

2. Relation to prior work

This paper makes a deliberate move that should be stated plainly before the formal sections begin: it is mostly a *reinterpretation* of well-established mathematical structures inside a single "ledger" vocabulary, not a wholesale replacement of those structures. The contribution is a unifying framework and a small number of derivations that follow from it; the contribution is not a new mathematics of complete positivity, entropy, horizon thermodynamics, or information geometry. This section positions each major claim of Paper 3 against the mainstream literature it draws on, and states explicitly where the paper is reinterpretive and where it is genuinely additive.

2.1 CPTP dynamics: Stinespring–Choi–Lindblad–GKSL

The result that admissibility-preserving evolution is exactly CPTP (T_{CPTP}, §4) lives in the orbit of the Stinespring dilation theorem [5], the Choi characterization of CP linear maps in finite dimensions [6], and the Lindblad/GKSL generators of completely positive dynamical semigroups [7, 8]. These are canonical results dating to 1955–1976; the operator-sum/Kraus form is standard [9].

Paper 3's contribution here is *provenance*, not endpoint. The CPTP class is unchanged. What changes is the direction of the derivation: Stinespring proves that every CPTP map admits a dilation; Paper 3 instead derives the CPTP requirement from A1 + tensor-product structure,

with Stinespring then supplying the Kraus form. The reader should not expect T_{CPTP} to deliver a class of maps outside CPTP, or to supersede Choi's classification; it should be read as a structural account of why the CPTP condition holds in this framework. The Technical Supplement §2 makes the comparison explicit through three differences (derivation direction, physical interpretation of the dilation, and finite-dimensional restriction).

2.2 Entropy and the second law: Lieb–Ruskai, Lindblad, Spohn

The identification of von Neumann entropy with committed capacity (T_{entropy}, §6) and the structural derivation of the second law ($T_{\mathrm{second\,law}}$, §7) draw on three lines of established work. Lieb–Ruskai strong subadditivity [10] is the entropy-inequality backbone for monotonicity and data-processing arguments throughout the paper; Wehrl's review [13] is the standard reference for general entropy properties. Lindblad's CP-map entropy inequalities [7] and Spohn's entropy-production framework for quantum dynamical semigroups [11] are the closest mainstream comparators for the second-law claim.

Paper 3's reframing here is rhetorical and ontological rather than mathematical: irreversibility is identified with local unrecoverability of cross-interface correlations rather than with semigroup thermodynamics. The outcome — positive entropy production along admissible trajectories — is the same as Spohn's; the route to it is different. The paper does *not* present a new entropy inequality of comparable mathematical independence to Lieb–Ruskai. What is added is the ledger-accounting principle (§6.1, formalized as a proposition in Technical Supplement §4): the entropy count is robust under small cost-landscape perturbations because counts do not depend on cost details.

2.3 Horizon entropy: Bekenstein, Hawking, Gibbons–Hawking, Jacobson

The geometric carrying-capacity extension (§8) and the recovery of horizon entropy $S_H = A/4$ via the $1/2 \times 1/2$ bookkeeping live in the lineage of Bekenstein's original information-theoretic black-hole entropy [14], Hawking's semiclassical particle-creation result that established horizon thermodynamics as a fact rather than an analogy [15], the Gibbons–Hawking extension to cosmological event horizons in positive-Λ spacetimes [16], and Jacobson's derivation of the Einstein equation as an equation of state from horizon thermodynamics [17]. The holographic principle of 't Hooft and Susskind [18, 19] sits in the background as the area-law commitment.

Paper 3's contribution here is *combinatorial accounting*, not new gravitational physics. The $S = A/4$ area law is unchanged; the $1/2 \times 1/2$ decomposition (relational symmetrization $\times$ Hamiltonian-constraint redundancy, formalized in Technical Supplement §6 under hypothesis H2) is a specific bookkeeping account of where the standard 1/4 prefactor comes from. The claim that thermodynamic entropy, entanglement entropy, and horizon entropy are three specializations of one ACC-saturation count is a unifying re-reading; it does not predict horizon phenomena outside the Bekenstein–Hawking–Gibbons–Hawking framework. The de Sitter combinatorial value $S_{dS} = 61 \ln 102 \approx 282.12$ nats is specifically a Gibbons–Hawking-style cosmological-horizon entropy, with the $C_{\mathrm{total}} = 61$ count inherited from Paper 2 rather than derived in Paper 3.

2.4 Fisher information and the renormalization group: Braunstein–Caves, Petz, Bény–Osborne

The Fisher-information section (§12) draws on the quantum Fisher metric of Braunstein and Caves [20], the classification of monotone metrics by Petz [21], and the foundational information-

geometry framework of Amari and Nagaoka [22]. The closest external support for the "RG flow as Fisher gradient descent" claim is the program of Bény and Osborne [23], who formulate RG explicitly as a family of quantum channels with an induced information geometry on the space of admissible effective descriptions.

Paper 3's contribution here is the capacity-count normalization ($S = (d_{\text{eff}}/2) \ln \det G$ with $d_{\text{eff}} = 102$ from Paper 2) and the suggestion that this same Fisher gradient feeds the channel-crossing structure of Paper 4's flavor sector. The information-geometric RG program itself is not new to this paper; the specific embedding of APF capacity counting into that program is.

2.5 Reconstruction programs: Hardy, CDP, Masanes–Müller, Dakić–Brukner

This paper sits stylistically in the family of operational quantum reconstruction programs — Hardy's five reasonable axioms [24], the informational derivation of Chiribella, D'Ariano, and Perinotti [25], the derivation of Masanes and Müller [26], and the entanglement-special argument of Dakić and Brukner [27]. Compared to these programs, the APF inventory (PLEC's four components A1, MD, A2, BW plus FD1–FD6 constitutive principles, taken from Paper 1) is comparable to or lighter than peer inventories. The departure from the reconstruction tradition is scope: the APF program attempts not just quantum kinematics but also thermodynamics, horizons, flavor, and cosmology. That broader ambition is both this paper's distinctive feature and its largest source of proof-burden.

2.6 Arrow of time: Maccone, Caticha, Chen, Di Biagio, Buchholz

The structural account of the arrow of time (§§7–10, especially the rebuttal of the typicality framing in §10.3) lands in a contested foundational literature. Several non-equivalent recent treatments offer alternative escapes from the typicality framing: Maccone's information-theoretic unrecordability [31]; Caticha's entropic-dynamics inference [32]; Chen's initial-state-realism approach [33]; the operational-asymmetry framework of Di Biagio, Donà, and Rovelli [34]; the algebraic-QFT framework of Buchholz and Yngvason [35]. Paper 3's resolution is constraint-on-the-admissible-set: low initial entropy is reframed as a small initial admissible set whose size is computable from A1, not as one improbable draw from a uniform measure. This is closer in spirit to Caticha and Chen than to Maccone's unrecordability or Buchholz's algebraic-QFT framing.

2.7 CP violation: Jarlskog and the empirical anchor at T2K

The CP-violation section (§13) proposes two structurally distinct mechanisms: holonomy phases (CKM-like) and entropy-optimization phases (PMNS-like). The mainstream structural treatment of CP violation is the invariant-CP program of Jarlskog [28, 29], which phrases CP-violation in basis-independent terms via invariants of the mass matrices. The two-mechanism split offered here is not Jarlskog's result; it belongs to the same broader effort to phrase CP-violation basis-independently. Where Jarlskog gives invariant algebraic structures, Paper 3 offers geometric-holonomy and entropy-extremization mechanisms.

The numerical phases ($\delta_{\text{CKM}} \approx 1.16$ rad, δ_{PMNS} specific value) are deferred to Paper 4. The empirical anchor that any sequel claim about δ_{PMNS} will ultimately have to meet is the long-baseline neutrino-oscillation programme, of which the T2K Collaboration's 2020 *Nature* measurement of δ_{CP} [30] is currently the strongest published constraint. The structural treatment in this paper is therefore not free-floating; it has a real target surface.

2.8 What this section does not do

This section positions Paper 3 in the established literature. It does not claim that Paper 3 supersedes any of the works cited. Specifically:

- T_{CPTP} is a reinterpretation of standard CP-map theory, not a replacement.
- T_{entropy} uses the standard Lieb–Ruskai monotonicity backbone; the contribution is the cost-interpretation, not a new entropy inequality.
- $S_H = A/4$ is the standard Bekenstein–Hawking–Gibbons–Hawking area law; the contribution is the $1/2 \times 1/2$ bookkeeping decomposition.
- RG-as-Fisher-descent is directionally aligned with Bény–Osborne; the contribution is the capacity-count normalization.
- The two-mechanism CP framing belongs in conversation with Jarlskog's invariant-CP program; the contribution is the structural separation, not the invariant structure.

The reader should encounter the formal sections that follow with this context in mind. The framework is unifying; the constituent results are mostly familiar; the genuine novelty lies in their joint reorganization under finite-enforcement bookkeeping.

3. Part I: Admissible Dynamics

Operational primitives: where each ledger term is defined and used

Paper 3 takes Paper 1's static admissibility structure (A1 + PLEC + tensor products T_{tensor}) and Paper 2's capacity counting ($C_{\text{total}} = 61$, $d_{\text{eff}} = 102$) as inputs. Five operational primitives carry the dynamical and entropic argument that follows. This box is a wayfinder, not new content; every claim points to where the term is defined or formalized.

Ledger. The bookkeeping of committed enforcement cost across an interface. "Committed" means capacity that cannot be locally reclaimed once spent. *Used as organizing metaphor throughout; formalized in two specific ways:* (i) entropy as committed-capacity count (T_{entropy}, §6); (ii) ACC as the geometric specialization (§8). *Foundational discussion:* §2.

CPTP map. An admissibility-preserving evolution. *Derived as theorem (not definition):* T_{CPTP} (§4); standard mathematical content from Stinespring–Choi–Lindblad–GKSL is reinterpreted, not replaced (see §2).

Committed capacity / cross-interface correlation. Capacity that has been allocated to a correlation across two interfaces. Once committed, it cannot be reclaimed by any local operation at one interface. *Formalized:* L_{irr} (§10). *Quantitative form:* $\kappa = 2$ multiplier of T_κ (§6); entropy as the count of such commitments via T_{entropy} (§6).

Admissibility carrying capacity (ACC). The general-interface capacity concept of which enforcement cost is the quantum-interface specialization. An ACC interface may be a quantum bipartition, a causal horizon, or a cosmological boundary. *Defined and developed:* §8. *Specializations:* (i) horizon entropy $S = A/4$ via the

$1/2 \times 1/2$ bookkeeping; (ii) de Sitter $S_{dS} = 61 \ln 102$ via C_{total} and d_{eff} from Paper 2.

Constraint functional Φ_c. The dimensionless excess between unconstrained and admissible mode counts: $\Phi_c = \ln(d_{\text{field}}/d_{\text{admissible}})$. Monotone-non-increasing along admissible cosmological trajectories under hypothesis H3 (Technical Supplement). *Defined:* §11. *Used:* the staged-relaxation cosmogenesis picture (§11); the macroscopic-arrow argument (§10, closing).

The Technical Supplement provides full formal treatments and the operational protocol for measuring committed capacity from physical observables.

4. Complete Positivity as a Theorem (T_{CPTP})

Paper 1 established that admissibility-preserving maps must be trace-preserving and positive: the trace is committed capacity normalized to one, and maps that fail positivity produce states with negative enforcement costs, which are inadmissible. What Paper 1 left conditional was the step from positivity to *complete* positivity.

The distinction matters. A map Φ on a system S is positive if it maps valid density matrices to valid density matrices. It is completely positive if $\Phi \otimes \mathbb{1}_R$ is also positive for any ancilla system R evolved by the identity map. The difference is ancilla-stability: a merely positive (but not completely positive) map can produce unphysical states when the system is entangled with a reference system that does not evolve.

Paper 1 deferred this step because ancilla-stability requires the tensor product structure (T_{tensor}) to be in place, and requires an analysis of how admissible maps compose when the ancilla carries its own enforcement budget. Both prerequisites are now available.

Theorem T_{CPTP} (Complete Positivity) [P]

Statement. Let $\Phi : \mathcal{B}(\mathcal{H}_S) \to \mathcal{B}(\mathcal{H}_S)$ be any map that preserves admissibility of the enforcement regime at interface Γ_S. Then Φ is completely positive and trace-preserving (CPTP), and admits a Kraus decomposition

$$\Phi(\rho) = \sum_k K_k \, \rho \, K_k^\dagger, \qquad \sum_k K_k^\dagger K_k = \mathbb{1}.$$

Variables:

- $\mathcal{H}_S$ — Hilbert space of the system at interface Γ_S
- K_k — Kraus operators; k indexes the environment outcomes
- ρ — density matrix of the system

Proof (three steps).

Step 1 (Trace preservation). A1 fixes $C(\Gamma_S)$ as a finite, positive constant: the enforcement budget at interface Γ_S. The trace $\operatorname{tr}\rho = 1$ is the normalization of that budget ($C_{\text{active}}/C(\Gamma_S) = 1$). An admissibility-preserving map redistributes committed capacity among states but cannot alter $C(\Gamma_S)$ itself — doing so would change the enforcement budget, which is precisely the datum A1 fixes. Hence:

- If $\operatorname{tr}\Phi(\rho) > 1$: capacity has been created. A1 violation (budget exceeds its fixed value).

- If $\operatorname{tr}\Phi(\rho) < 1$: capacity has been annihilated. A1 violation equally (budget falls below its fixed value; A1 requires positivity *and* conservation of the budget, not merely an upper bound).

Therefore $\operatorname{tr}\Phi(\rho) = 1$ unconditionally, with A1 as the sole source. (L_{irr} is not needed here; it enters only in Step 2.)

Step 2 (Complete positivity via ancilla-stability). By T_{tensor} (Paper 1), the joint system $S \otimes R$ for any ancilla R carries an independent enforcement budget $C(\Gamma_R)$. The state ρ_{SR} is admissible if and only if it has non-negative enforcement costs at all interfaces. Now apply $\Phi \otimes \mathbb{1}_R$: Φ acts on Γ_S and the identity acts on Γ_R. Since Γ_R is independent of Γ_S (by L_{loc}), the evolution at Γ_S cannot create inadmissible states at Γ_R. The joint map therefore preserves admissibility of ρ_{SR} for any initial $\rho_{SR} \geq 0$. This is the definition of complete positivity.

Step 3 (Kraus representation). Given that Φ is CPTP, the Stinespring dilation theorem [5] applies: there exists a Hilbert space $\mathcal{H}_E$ (the environment) and an isometry $V : \mathcal{H}_S \to \mathcal{H}_S \otimes \mathcal{H}_E$ such that $\Phi(\rho) = \operatorname{tr}_E[V\rho V^\dagger]$. The Kraus operators [6, 9] are $K_k = (\mathbb{1}_S \otimes \langle k|_E)V$ for any orthonormal basis $\{|k\rangle\}$ of $\mathcal{H}_E$.

Enforcement interpretation of the dilation. The Stinespring isometry V describes the system and environment evolving unitarily, with each maintaining a separate enforcement budget. The partial trace over E discards the environmental budget: L_{irr} ensures that the capacity committed to S–E correlations cannot be recovered at Γ_S, so tracing out E is not an approximation but the exact enforcement account.

Code reference: `check_T_CPTP` in `core.py`.

Remark on Paper 1's conditional statement. Paper 1 listed T_{CPTP} as conditional on the identification "ancilla-stability: coupling to an ancilla preserves admissibility." That identification is now a theorem: Step 2 above derives it from T_{tensor} and L_{loc}. The conditional flag on T_{CPTP} in Paper 1's Table H.1 is accordingly lifted.

What T_{CPTP} does not say. The theorem characterizes the *class* of admissible evolution maps. It does not specify which map is selected in any given physical situation; that requires the dynamical structure developed in Paper 6 (Euler-Lagrange and geodesic equations from enforcement cost extremization).

4.1 Sequential composition and the no-bit-commitment corollary

Theorem T_{seq} (Sequential Admissibility) [P]

Statement. Composites of admissible CPTP maps are admissible CPTP maps. For T time steps, the cumulative enforcement cost is bounded:

$$\sum_{t=1}^{T} \varepsilon(d_t) \;\leq\; C(\Gamma) \cdot T.$$

Time-evolved observables $a(t) = U_t^\dagger \, a \, U_t$ satisfy the same Cirel'son operator identity as static observables, giving the Tsirelson bound $2\sqrt{2}$ for temporal CHSH correlators.

Proof. CPTP maps form a semigroup under composition: the composite of two trace-preserving maps is trace-preserving, and the composite of two completely positive maps

is completely positive (tensor product of Kraus operators). This is the standard GKSL–Lindblad picture [7, 8] interpreted via the enforcement ledger. The cost bound follows because each step draws at most $\varepsilon(d_t) \leq C(\Gamma)$ from the budget; the total over T steps is bounded by $C(\Gamma) \cdot T$ by subadditivity.
For the dynamic Tsirelson bound: the Cirel'son identity $\|[a(t_1), a(t_2)]\| \leq 2$ holds for any two unitary-evolved ± 1 observables because unitary conjugation is an automorphism of the operator algebra, and the norm bound $\|[a, b]\| \leq 2\|a\|\|b\|$ is algebraic, not state-dependent. The Tsirelson bound for temporal correlators follows from the same Cirel'son identity as for static ones.

Corollary block: no-cloning, no-deleting, no-bit-commitment.

Three constraints on information processing follow immediately from T_{CPTP} and the capacity accounting of $C(\Gamma) = 61$.

No-cloning. A perfect cloning map would send $\rho \mapsto \rho \otimes \rho$. This doubles the enforcement demand at Γ_S: the output requires $2\,C(\Gamma)$ to maintain. Since A1 fixes $C(\Gamma)$, the map is inadmissible. More formally, a cloning map cannot be CPTP because it is not linear: for $\rho = p\rho_1 + (1-p)\rho_2$, the cloning map gives $\rho^{\otimes 2}$, which differs from $p\rho_1^{\otimes 2} + (1-p)\rho_2^{\otimes 2}$ unless the states are orthogonal. CPTP maps are linear; cloning is not.

No-deleting. Deletion of one copy from $\rho \otimes \rho$ would require a map that reduces the environmental record at Γ_E. But L_{irr} prohibits local erasure of cross-interface correlations. The deletion map is inadmissible.

No-bit-commitment. A bit-commitment protocol requires: (i) *hiding* — the receiver cannot determine the committed bit b from the committed state ρ_b; and (ii) *binding* — the committer cannot change b after commitment. These two requirements are jointly inadmissible. If hiding holds then $\rho_0^R = \rho_1^R$ (the receiver's reduced state is identical for both bits). By Uhlmann's theorem plus the Stinespring dilation supplied by T_{CPTP}, the joint purified states satisfy $|\psi_1\rangle = (U_C \otimes \mathbb{1}_R)|\psi_0\rangle$ for some unitary U_C on the committer's system. But U_C is a CPTP map (unitary evolution is a special case), so the committer can apply it after commitment, transforming ρ_0 into ρ_1: binding fails. Hence hiding $\Rightarrow$ not binding, and no protocol achieves both simultaneously.

5. Part II: The Ledger

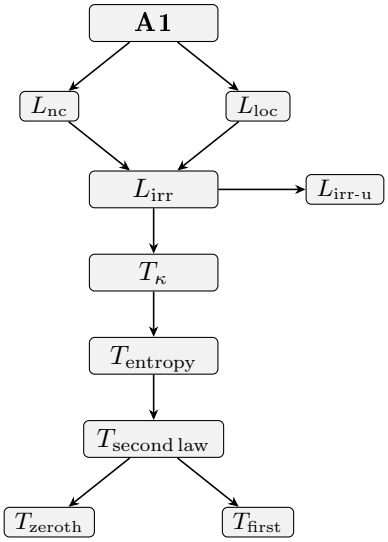
6. Entropy as Committed Capacity (T_{entropy}, T_κ)

Paper 1 introduced the von Neumann entropy functional [12] as a consequence of the operator algebra, but left its physical interpretation informal: entropy was described as measuring "how much has been committed." This section makes that identification precise, derives the directed-enforcement multiplier $\kappa = 2$ that controls the cost of each commitment, and shows that the standard laws of thermodynamics follow immediately. The strong-subadditivity infrastructure that supports monotonicity and data-processing arguments throughout what follows is the Lieb–Ruskai backbone [10, 13].

The approach is bottom-up. We first establish that the cost per distinction cannot be $\kappa = 1$ (single-interface commitment is erasable) and cannot be $\kappa > 2$ (a third independent interface

commitment would require a second environment, which is a new correlation, not an additional cost on the original one). The unique value is $\kappa = 2$.

The derivation chain for this section is shown below.



Theorem T_κ (Directed Enforcement Multiplier) [P]

Statement. The directed enforcement multiplier is $\kappa = 2$. Each maintained distinction incurs exactly two independent interface commitments: one at the system interface Γ_S (forward stabilization) and one at the environment interface Γ_E (environmental record).

Proof.

Lower bound $\kappa \geq 2$. Step 1: L_{nc} requires forward enforcement at Γ_S — without active stabilization, distinctions collapse. This commits at least ε at Γ_S per distinction. Step 2: L_{irr} requires that when a system creates a distinction, the S–E correlation ($\Delta > 0$) commits capacity at Γ_E independently. This is the environmental record. Step 3: by L_{loc}, Γ_S and Γ_E are independent budgets. Freeing C_{fwd} at Γ_S does not free C_{env} at Γ_E. The two commitments are structurally distinct. Total cost $\geq 2\varepsilon$ per distinction, so $\kappa \geq 2$.

Upper bound $\kappa \leq 2$. A third commitment would require a third independent interface — a second environment. But a second environment constitutes a new distinction (the cross-interface correlation between the two environments), not an additional enforcement obligation on the original distinction. PLEC's argmin component (A2, the minimum-cost selection feature of the admissibility variational principle — enforce exactly what is needed to maintain admissibility, no more) then gives $\kappa \leq 2$.

Uniqueness. $\kappa \geq 2$ and $\kappa \leq 2$ together give $\kappa = 2$.

Physical interpretation. $\kappa = 2$ is the enforcement analogue of the Nyquist sampling theorem: two independent samples (system and environment) are required to fully characterize a distinction's enforcement state. The environment is the independent auditor.

Code reference: `check_T_kappa` in `core.py`.

With $\kappa = 2$ in hand, the identification of entropy with committed capacity becomes sharp.

Theorem T_{entropy} (Von Neumann Entropy as Committed Capacity) [P]

Statement. The entropy $S(\Gamma, t) = \mathcal{E}_\Gamma(R_{\text{active}}(t))$, defined as the enforcement demand of active correlations at interface Γ, equals the von Neumann entropy $S(\rho) = -\text{tr}(\rho \log \rho)$ in any quantum-admissible regime.

The entropy satisfies five properties, all derived from capacity structure:

1. $S \geq 0$ (enforcement costs are non-negative).
2. $S = 0$ if and only if ρ is pure (no committed capacity beyond the minimum).
3. $S \leq \log d$ with equality at maximal mixing (capacity saturation).
4. Subadditivity: $S(AB) \leq S(A) + S(B)$ (non-closure bounds joint enforcement).
5. Concavity: $S(\sum_i p_i \rho_i) \geq \sum_i p_i S(\rho_i)$ (mixing never decreases committed capacity).

Proof. The identification $S = -\text{tr}(\rho \log \rho)$ follows from the uniqueness of the capacity cost function $F(d) = d$ (proved via Cauchy's functional equation in Paper 2, L_{Cauchy}) together with the spectral decomposition $\rho = \sum_i \lambda_i |i\rangle\langle i|$: the cost of maintaining n orthogonal distinctions weighted by λ_i is $-\sum_i \lambda_i \log \lambda_i = -\text{tr}(\rho \log \rho)$. Properties 1–3 are direct from capacity non-negativity and the $[0, \log d]$ range of committed fractions. Property 4 (subadditivity) follows from L_{nc}: non-closure means joint enforcement at a shared interface exceeds individual costs, but the excess $\Delta > 0$ forces $S(AB) \leq S(A) + S(B)$ via the data-processing inequality. Property 5 (concavity) follows because mixing increases the number of distinctions being simultaneously maintained.

Code reference: `check_T_entropy` in `core.py`.

6.1 The ledger-accounting principle

The entropy just derived is a *count* of the microscopic configurations consistent with the ledger's cost constraints—nothing more. This is a deep structural feature, not a convenience. The multiplicity of admissible configurations depends on the cardinality of the admissible set, not on the detailed cost landscape that defines it. Two consequences follow.

First, interaction corrections that shift the energy of the saturated configuration leave the configuration count invariant: interactions modify which state is lowest-cost, but the count of admissible states above that minimum is unchanged. Second, thermal corrections that alter the relative weights of states with different costs also leave the saturated count invariant, because at saturation every admissible configuration has been realized and there are no unrealized states to reweight.

This is the key to the exactness of the cosmological entropy bound: the de Sitter entropy $S_{dS} = C_{\text{total}} \ln d_{\text{eff}} = 61 \ln 102$ is not a leading-order approximation refined by interactions; it is a combinatorial identity in the admissible set. The cosmological horizon to which this identity attaches is the Gibbons–Hawking horizon [16] of a positive-Λ de Sitter spacetime; the combinatorial value $61 \ln 102$ is interpreted as the ACC-saturation entropy of that horizon, with the count $C_{\text{total}} = 61$ inherited from Paper 2's capacity-partitioning argument. This is therefore a structural cosmological-entropy claim within the Bekenstein–Hawking–Gibbons–Hawking framework, not a competitor to it. The thermodynamic formalism (partition function, finite-temperature corrections to the free energy, beta-dependence of individual observables) is developed in Paper 7; Paper 3 needs only the accounting principle itself. We return to the point in §8.4, where the same accounting unifies thermodynamic, entanglement, and horizon entropy as three views of interface-capacity saturation.

6.2 Thermodynamic laws as corollaries

The zeroth and first laws of thermodynamics follow from T_{entropy} and L_{irr} without additional assumptions.

Theorem T_{zeroth} (Zeroth Law) [P]

Statement. Define the inverse temperature $\beta = \log d/\varepsilon$ and the temperature $T = \varepsilon/\log d$. Capacity flows spontaneously from high-T (low-β) to low-T (high-β) interfaces. Equilibrium is the state $\beta_1 = \beta_2$, and this condition is transitive: if Γ_1 is in equilibrium with Γ_2, and Γ_2 with Γ_3, then Γ_1 is in equilibrium with Γ_3.

Derivation. Transfer of one unit from Γ_2 to Γ_1 changes entropy by $\Delta S = (\beta_1 - \beta_2)\varepsilon$. By L_{irr}, entropy-increasing transfers are spontaneous; flow stops when $\beta_1 = \beta_2$. Transitivity is transitivity of real-number equality. The cosmological equilibrium temperature is $T_{\text{univ}} = \varepsilon/\log 102$.

Code reference: `check_T_zeroth_law` in `supplements.py`.

Theorem T_{first} (First Law) [P]

Statement. Energy changes at any interface decompose as $dE = dQ + dW$, where $dQ = T\,dS$ (heat: irreversible, entropy-producing, forced by L_{irr}) and $dW = dE - T\,dS$ (work: reversible, entropy-neutral).

Derivation. L_{irr} partitions any dE into irreversible (commitment-generating) and reversible (commitment-neutral) components. Pure commitment: $dQ = T\,dS$, $dW = 0$. Pure work: $dQ = 0$, $dW = dE$. The mixed case is the accounting identity $dE = dQ + dW$. For the cosmological fill through all 61 capacity types, every commitment is pure heat ($dW_{\text{total}} = 0$): the universe does no work on itself during the capacity fill.

Code reference: `check_T_first_law` in `supplements.py`.

7. The Second Law ($T_{\text{second law}}$)

The second law is the claim that entropy does not decrease. In statistical mechanics this is usually derived from typicality arguments or taken as a boundary condition on the initial state. The closest operator-algebraic treatment is Spohn's positive-entropy-production framework for quantum dynamical semigroups [11], which establishes approach to equilibrium from completely positive evolution and is the mainstream comparator for the result below. In the admissibility framework the second law is derived not from semigroup thermodynamics but from L_{irr}: the local unrecoverability of cross-interface correlations. The outcome — positive entropy production along admissible trajectories — agrees with Spohn; the route is different, and the substantive novelty is the rhetorical shift rather than a replacement of the monotonicity result itself (see §2).

Theorem $T_{\text{second law}}$ (Second Law of Thermodynamics) [P]

Statement. The entropy of any closed subsystem is non-decreasing under admissibility-preserving evolution. The entropy of the universe is strictly increasing during the capacity fill and constant at saturation.

Level A: Subsystem second law.
For any CPTP map Φ arising from tracing over an environment that begins in a low-entropy state:

$$S(\Phi(\rho_S)) \;\geq\; S(\rho_S).$$

Proof. By T_{tensor}, the total system-environment evolution is unitary: $S(\rho_{SE}(t)) = S(\rho_{SE}(0))$ (unitary evolution is entropy-preserving). By L_{irr}, the partial trace over E discards cross-interface correlations that cannot be recovered at Γ_S. Information about S that has leaked to E is gone from the perspective of any S-local observer. The data-processing inequality then gives $S(\Phi(\rho_S)) = S(\text{tr}_E[\rho_{SE}(t)]) \geq S(\rho_S)$ for unital channels; strict inequality holds whenever new correlations form.

Level B: Cosmological second law.
Define the k-step entropy $S(k) = k \ln d_{\text{eff}} = k \ln 102$ for $k \in \{0, 1, \ldots, 61\}$ capacity types committed. Then:

1. $S(k)$ is strictly increasing: $S(k+1) - S(k) = \ln 102 > 0$.
2. L_{irr} makes k non-decreasing in time: committed types cannot be uncommitted by any local operation.
3. Therefore $dS/dt \geq 0$ globally, with equality only at saturation $k = 61$.
4. At saturation: $S = S_{dS} = 61 \ln 102 \approx 282.12$ nats $= S_{\max}$.

Not a boundary condition. The standard derivation of the second law in statistical mechanics requires either a special low-entropy initial state (the "past hypothesis") or a Boltzmann-style typicality argument. Neither is needed here. L_{irr} gives irreversibility as a structural consequence of A1, L_{nc}, and L_{loc} without any assumption about initial conditions. The capacity fill begins at $k = 0$ (the empty ledger) and proceeds to $k = 61$ because each commitment is irreversible. The initial low-entropy state is not assumed; it is forced by the fact that no capacity had yet been committed.

Code reference: `check_T_second_law` in `supplements.py`.

8. Geometric Carrying Capacity

The ledger developed in §§ 6–7 tracks the cost of maintaining distinction across a *quantum* interface: a bipartition of a pure state into system and environment, or the interface between a system and its purifying ancilla. We now extend the framework to general relational interfaces, including causal and cosmological ones. The extension is natural: every interface carries an admissibility capacity—an upper bound on the number of independent relational degrees of freedom that can be coordinated across it—and saturation of this capacity is the universal condition under which entropy attains its characteristic scaling.

8.1 From enforcement cost to interface capacity

For a quantum interface, the capacity is the dimension of the relevant tensor factor and the cost is the enforcement cost ε^* per distinction maintained. For a causal interface—a horizon—or a cosmological boundary, the capacity is bounded by the geometric area of the interface in Planck units, and the cost is the enforcement cost of maintaining a distinction accessible at that interface. We call the general upper bound the *admissibility carrying capacity* (ACC) of the interface. The ACC is an upper bound, not a state; it plays the same role that enforcement capacity plays in Paper 1 and the same role that C_{total} plays in the cosmological bookkeeping of

Paper 2. What counts as "enforcement" is the maintenance of relational distinction across the interface; what counts as "saturation" is the condition that the number of admissible distinctions equals the ACC.

This extension is not a new axiom. It follows from A1 (finite capacity applies to every enforcement problem) and the observation that an interface with finite geometric extent can only support finitely many independent relational distinctions. The full derivation—showing that the geometric bound is Bekenstein area-law in structure—is developed in Paper 5 and Paper 6 (spacetime and gravity); here we take it as given and focus on the entropic consequences.

8.2 Relational bookkeeping: the two factors of one-half

Two factors of 1/2 must be applied to the naive ACC count when converting an interface area into an entropy. They have a principled origin and are not free parameters.

Factor 1: Relational double-counting. A distinction across an interface belongs to both sides. Counting it once from each side and then dividing by two is the only symmetric choice consistent with T_{seq} composition. Formally, this is the factor recovered when a two-point correlator is decomposed into its symmetric and antisymmetric pieces and only the independent half is kept.

Factor 2: Gauge redundancy at the interface. The admissibility constraints themselves are maintained by the same degrees of freedom they constrain; half of the interface modes are redundant with the enforcement of the constraint itself. This is the Hamiltonian-constraint factor familiar from canonical gravity: in a theory whose Hamiltonian vanishes on constraints, only half of the boundary degrees of freedom are independent.

Together these give the interface entropy

$$S_{\text{interface}} = \tfrac{1}{4}\,\frac{A}{\ell_P^2}\cdot\frac{\ln d_{\text{eff}}}{\ln 2},$$

when the ACC is expressed in capacity modes of dimension $d_{\text{eff}} = 102$ rather than in bits. Here A is the area of the interface in the enforcement metric and ℓ_P is the Planck length. The leading 1/4 is the product of the two bookkeeping factors; the $\ln d_{\text{eff}}/\ln 2$ factor converts from capacity modes to bits to match standard conventions. Recovering the Bekenstein–Hawking form [14, 15] $S = A/4$ (in natural units) is automatic once the ACC-to-bit conversion is chosen conventionally. Both halves are forced by structure—symmetrization of relational counts and gauge-redundancy elimination—not fitted. The resulting area scaling is the same area law that the holographic principle [18, 19] and Jacobson's Einstein-equation-from-thermodynamics argument [17] each use as a starting point; here the area law is the downstream consequence, not the input.

8.3 Horizon entropy as saturation

Apply §8.2 to a causal horizon. The interface area A is the horizon area. The saturation condition—ACC fully used by realized distinctions—is the condition under which the horizon admits a thermal description: every admissible mode at the interface is occupied exactly as permitted by the ACC. Horizon entropy is therefore a special case of second-law saturation (§7), applied not to a quantum bipartition but to a geometric causal boundary.

This reframes two long-standing puzzles. First, horizon entropy has always been anomalous in standard statistical mechanics because it scales with area rather than volume; in the ACC picture, area scaling is mandatory because capacity is an interface quantity, not a bulk one.

Second, horizon entropy is universal across horizon types (black-hole, cosmological, Rindler); in the ACC picture this is because all three are cases of the same geometric capacity saturation.

Let S_{dS} denote the cosmological de Sitter entropy. The derivation in §7 ($S_{dS} = C_{\text{total}} \ln d_{\text{eff}} = 61 \ln 102 \approx 282.12$ nats) is now recovered as a special case of §8.2–8.3: the cosmological horizon is an ACC-saturated interface, the capacity-mode count is the structurally rigid $C_{\text{total}} = 61$ of Paper 2, and the per-mode dimension is $d_{\text{eff}} = 102$. The two derivations—combinatorial (§7) and geometric (§8)—agree because they are the same derivation viewed from two sides of the interface.

Code reference: `check_T_deSitter_entropy` in `gravity.py`.

8.4 Three entropies, one accounting

The ledger-accounting principle of §6.1 together with the interface framework of §8.1–8.3 yields a unification. Thermodynamic entropy, entanglement entropy, and horizon entropy are three specializations of a single quantity—the count of admissible configurations at capacity saturation of a relational interface.

- **Thermodynamic entropy.** The interface is the boundary between the system and an implicit coarse-graining or environment. Capacity is set by the system's admissible phase space; the count is the number of microstates consistent with the macroscopic constraints. Saturation recovers equilibrium.
- **Entanglement entropy.** The interface is the bipartition of a joint quantum state. Capacity is the Schmidt rank; the count is the Schmidt spectrum multiplicity at maximal mixing. Saturation recovers the entanglement bound.
- **Horizon entropy.** The interface is a causal or cosmological boundary. Capacity is the geometric ACC; the count is the number of admissible interface distinctions. Saturation recovers the Bekenstein–Hawking area law with prefactor 1/4.

No axiom beyond A1 and T_{entropy} is needed. What varies across the three cases is the interface; what is constant is the counting structure. The unification is a direct consequence of the ledger-accounting principle: entropy depends on the admissible set, not on the cost-landscape machinery used to define it.

9. Part III: Time, Cosmogenesis, Information Geometry, and CP Structure

10. The Arrow of Time (L_{irr}, $L_{\text{irr-uniform}}$, T_{CPT})

The second law shows that entropy does not decrease. The arrow of time is the sharper claim: that there is a preferred orientation on the single timelike dimension, and that this orientation is the direction of entropy increase. This section derives that claim and identifies the quantity that measures the asymmetry.

Lemma L_{irr} (Irreversibility from Admissibility Physics) [P]

Statement (mechanism: locally unrecoverable correlations). Let S and E be systems at independent interfaces Γ_S and Γ_E. After any interaction that creates cross-interface correlations $\Delta(S, E) > 0$, the pre-interaction state is unrecoverable by any

S-local operation.

Proof (four steps).

Step 1 (L_{nc}). L_{nc} gives $\Delta > 0$: the joint enforcement cost at the shared interface exceeds the sum of individual costs.

Step 2 (L_{loc}). Enforcement is local: the observer at Γ_S has no access to Γ_E, and vice versa.

Step 3 (Locally unrecoverable capacity). The interaction commits capacity $\Delta > 0$ at *both* Γ_S and Γ_E simultaneously. Freeing this capacity requires coordinated action at both interfaces. No local observer can perform this (Step 2). From S's perspective, the correlation capacity is permanently committed.

Step 4 (Structural irreversibility). Locally unrecoverable capacity = irreversibility: the pre-interaction state at Γ_S is unrecoverable by any S-local CPTP map. This is structural, not probabilistic.

Key distinction from entropy-lock arguments. Earlier treatments (APF v4.x) attempted irreversibility via a "record-lock": removing a distinction from a state would activate a conflict making the result inadmissible. This argument is invalid: if the enforcement demand $\mathcal{E}$ is monotone (as required by A1), then $S' \subset S$ implies $\mathcal{E}(S') \leq \mathcal{E}(S) \leq C(\Gamma)$, so the reduced state is always admissible. The present argument avoids this: all states remain globally admissible, but cross-interface capacity cannot be freed by any local operation. Monotonicity holds at each interface; irreversibility comes from limited access, not from inaccessible states.

Code reference: `check_L_irr` in `core.py`.

Lemma $L_{\text{irr-uniform}}$ (Sector-Uniform Irreversibility) [P]

Statement. L_{irr} applies uniformly to all 61 enforcement types, including the gauge-matter sector established in Paper 2, without additional assumptions.

Proof. The mechanism in L_{irr} uses only A1, L_{nc}, and L_{loc}. None of these distinguish between gauge channels and matter channels. By $T_{\text{irr-uniform}}$ (gauge sector has same capacity structure as matter sector, from Paper 2's T_{7B}), the argument applies uniformly. In particular, the chirality argument of Paper 2 (R2) can invoke L_{irr} in the gauge sector without additional justification.

Code reference: `check_L_irr_uniform` in `core.py`.

10.1 The arrow: orientation on the timelike direction

Paper 5 (Spacetime and Gravity) derives the Lorentzian signature $(-,+,+,+)$ from enforcement structure, establishing that there is exactly one timelike direction. L_{irr} selects an *orientation* on that direction: the forward direction is the direction of capacity commitment, which is the direction of non-decreasing entropy.

This gives the arrow of time a precise derivation: it is not a boundary condition, not a consequence of low-entropy initial data, and not a probabilistic tendency. It is the direction forced by the structural irreversibility of cross-interface correlations.

Theorem T_{CPT} (CPT Invariance and T-Violation) [P]

Statement. CPT is an exact symmetry of the admissible dynamics. T is violated by phase $\varphi_T = \pi/4$ (maximal). The T-violation quantifies the asymmetry between the forward and backward time directions.

Derivation.

1. $L_{\mathrm{irr}} \Rightarrow T$ broken: capacity commitment is irreversible, so the forward and backward time directions are not equivalent.
2. $B1' \Rightarrow C$ and P broken: the chiral gauge structure of the Standard Model (proved in Paper 2) breaks both parity and charge conjugation.
3. $L_{\text{holonomy phase}} \Rightarrow CP$ broken by phase $\pi/4$: the non-abelian holonomy phase from the SU(2) gauge structure contributes phase $\pi/4$.
4. $T_{\mathrm{CPT}} \Rightarrow CPT$ exact (Jost theorem / Pauli-Jordan): the C^*-algebra established by Paper 1's T2 satisfies the hypotheses of the PCT theorem. CPT is therefore an exact symmetry.
5. Combining: CP violated by $\pi/4$ and CPT exact forces T violated by $\pi/4$. The $\sin(2\varphi_T) = \sin(\pi/2) = 1$ is maximal.

Code reference: `check_T_CPT` in `supplements.py`.

The T-violation phase $\pi/4$ is a derived quantity. It is not an input to the framework; it is the output of L_{irr}, $B1'$, $L_{\text{holonomy phase}}$, and the algebraic structure of Paper 1's T2.

10.2 Macroscopic arrow from constraint loss

The L_{irr} argument of §10 gives a *quantum-interface* arrow: admissible dynamics cannot uniformly undo distinguishability across all initial pairs, so time's arrow points in the direction of distinguishability non-recovery. This quantum arrow is fundamental but operates at the interface scale. A complementary *macroscopic* arrow operates at cosmological scale.

At cosmological scale, admissibility is maintained not by a single enforcement interface but by the *coordinated* admissibility of all interfaces in a causally connected region. A globally coordinated admissible state is one in which the admissibility constraints at every interface are simultaneously satisfied; an uncoordinated state is one in which admissibility holds locally but fails globally. Finite capacity implies that coordination across an expanding region cannot be maintained indefinitely: as the causal diamond grows, the joint ACC of all interfaces in the region eventually exceeds what finite enforcement can maintain, and coordination begins to fail.

The direction in which coordination can be lost but not regained defines the macroscopic arrow of time. This is not a new irreversibility lemma; it is L_{irr} applied to the coordinated-admissibility observable over a macroscopic region. It identifies the thermodynamic arrow with *loss of global admissible coordination*, not with asymmetry of microscopic laws. The microscopic dynamics remain CPTP and T_{CPT}-respecting, as §4 proved; the asymmetry is structural, arising from the finiteness of the enforcement budget available to maintain global coordination.

This macroscopic arrow is what couples to cosmology: the staged loss of coordination is the engine of cosmogenesis, treated in §11.

10.3 Low entropy means restricted phase space, not improbable order

A standing puzzle in statistical cosmology is why the universe began in a low-entropy state. The typicality framing—that the initial state was improbable among all compatible microstates—has

long felt unsatisfactory, because "improbable" presupposes a measure we do not have and a dynamics we did not choose. Recent foundational treatments offer several non-equivalent escapes from the typicality framing: Maccone's information-theoretic unrecordability [31], Caticha's entropic-dynamics inference [32], Chen's initial-state-realism approach [33], the operational-asymmetry framework of Di Biagio et al. [34], and the algebraic-QFT framing of Buchholz and Yngvason [35]. The admissibility framework's resolution is structural rather than statistical or operational, and is closer to the constraint-on-the-admissible-set spirit of Caticha and Chen than to the unrecordability spirit of Maccone.

The admissibility framework resolves the puzzle structurally. A low-entropy initial state is a state with *restricted admissible phase space*: fewer configurations were compatible with the admissibility constraints, because the early enforcement environment was overconstrained. Entropy is low not because the initial configuration was one improbable draw from a large uniform sample, but because the admissible set itself was small. The question "why was the initial entropy low?" becomes "why was the initial admissible phase space restricted?"—and that has a concrete structural answer (§11): because the early universe sat in an overconstrained regime where most gravitational degrees of freedom were inadmissible, and entropy grew as constraint was staged-relaxed.

This is not a redefinition. It is a change in the explanatory burden. Typicality arguments must posit an improbable initial condition; the constraint-functional framing (§11.1) posits a small initial admissible set whose size is computable from A1.

11. Cosmogenesis and Staged Relaxation

The macroscopic arrow of §10.2 and the restricted phase-space reframing of §10.3 together set up a structural treatment of cosmogenesis. This section develops it. No new axiom is introduced; the section is a consequence of A1 applied at cosmological scale, combined with the ledger accounting of §6 and the geometric carrying capacity of §8. Quantitative cosmogenesis—specific geometries, spectra, and observational predictions—requires the Paper 6 development of the cost landscape and is beyond the scope here.

11.1 The constraint functional

Define the *constraint functional* of a capacity budget:

$$\Phi_c \;=\; \ln\!\left(\frac{d_{\text{field}}}{d_{\text{admissible}}}\right),$$

where d_{field} is the unconstrained cardinality of the field configuration space and $d_{\text{admissible}}$ is the cardinality of the admissible subset. Φ_c measures how much of the naive field content is inadmissible in the current enforcement environment. A region with $\Phi_c = 0$ is fully permissive; a region with Φ_c large is overconstrained.

Two structural properties make Φ_c useful. First, Φ_c is *monotone non-increasing* in time under admissibility-preserving dynamics: once a constraint is relaxed, it cannot be restored by any CPTP evolution (§10, L_{irr}). Second, Φ_c sets a bound on the admissible entropy through

$$S_{\text{admissible}} \;\leq\; \ln d_{\text{field}} - \Phi_c,$$

so entropy increases precisely as the constraint functional decreases. This is the quantitative form of the qualitative claim in §10.3.

11.2 The overconstrained initial regime

The earliest admissible regime is the one with maximum Φ_c: nearly all field configurations are inadmissible, and only a small, highly constrained subset is realized. In gravitational terms, this is an overconstrained state in which most gravitational degrees of freedom cannot be dynamically realized because the admissibility constraints of neighboring interfaces conflict. This is not a boundary condition; it is the state of the enforcement environment in the deep pre-relaxation regime.

We emphasize what this framing does and does not claim. It *does* claim that the low initial entropy is structural—Φ_c is large because $d_{\text{admissible}}$ is small—and that staged relaxation naturally yields monotonic entropy growth. It *does not* claim a specific geometry for the overconstrained regime, a quantitative pre-big-bang cosmology, or an identification of $\Phi_{c,\text{initial}}$ with any particular observational quantity. This paper provides the structural framing; specific cosmological models belong to Paper 6.

11.3 Staged relaxation and scale-dependent imprints

Constraint relaxation cannot be uniform across scales. Different modes have different admissibility costs, and the mode with the lowest cost relaxes first; only once a given mode has saturated its share of the enforcement budget can the next mode begin to relax. The result is a *staged* relaxation process: constraints are released in sequence, not in parallel, and each stage imprints its characteristic scale on the resulting relic structure.

The consequence is a *scale-dependent hierarchy* in observables that probe the early universe. Modes that relaxed at similar stages share correlations; modes that relaxed at different stages do not. This predicts structured (not pure power-law) imprints in any observable sensitive to the pre-saturation epoch: primordial gravitational-wave spectra, late-time ringdown tails, and certain coherence measures.

This is not a quantitative model; it is a structural expectation. Testing it requires a model of the enforcement-cost landscape across modes, which Paper 6 develops. For present purposes the claim is structural: staged relaxation is a consequence of A1 plus bounded enforcement per stage, and its absence would falsify the admissibility framing of cosmogenesis (see F5 in §15).

11.4 The trapped-surface threshold

The boundary between the overconstrained regime and the staged-relaxation regime is the *trapped-surface threshold*: the locus in state space where the joint admissibility of all interface constraints in a region ceases to be enforceable. In geometric terms, this is where global causal structure would require the formation of a trapped surface—formally, where the expansion of every causal congruence becomes negative.

Inside the threshold, constraint cannot be locally dynamically enforced: the state is inadmissible. Outside, constraint is enforceable and dynamics proceeds. The threshold itself is a saturation boundary in the ACC sense of §8.1. This connects the cosmogenesis of Paper 3 to the horizon physics of §8.3: trapped surfaces are cosmological horizons, and their entropy obeys the same counting as any other ACC-saturated interface.

11.5 Horizon enforceability: a preview of Paper 6

An admissibility interface that saturates its ACC cannot simultaneously maintain (a) smooth classical geometry across the interface *and* (b) global unitarity of the effective dynamics. The

argument is structural: smooth geometry requires the interface to admit a continuum of local modes, and global unitarity requires all these modes to remain mutually distinguishable; but at ACC saturation, the interface has only finitely many independent distinctions to allocate, so one of (a) or (b) must give. *Smoothness fails before unitarity.*

This is not a new theorem proved here; Paper 6 develops it as the natural closure of the dynamics/geometry treatment. We preview it in Paper 3 because it is the natural end-point of the entropy-as-interface-saturation story: at saturation, the very description of the interface as a smooth geometric boundary breaks down, and what remains is the combinatorial ACC count—exactly the entropy of §8.3. The observable signatures (qualitative EFT breakdown near horizons, non-Markovian horizon memory, deviations in late-time ringdown) enter the falsifier list as F4 and F5 in §15.

12. Fisher Information and the Entropy Gradient

The entropy functional $S(\rho) = -\mathrm{tr}(\rho \log \rho)$ induces a natural Riemannian structure on the space of admissible states. This section shows that the induced metric is the Fisher information metric of Braunstein–Caves [20] – equivalently, the unique monotone metric up to normalization in the Petz classification [21, 22] – and that the renormalization group flow is Fisher gradient descent on this metric.

The starting point is the capacity-counting structure of Paper 2: the 61 enforcement types define a discrete parameter space, and the entropy $S(k) = k \ln 102$ defines a monotone function on it. When we pass to the continuous density-matrix description, this discrete counting becomes a smooth metric.

Lemma L_{Fisher} (Fisher Information from Capacity Counting) [P]

Statement. Capacity counting for $N_{\mathrm{gen}} = 3$ generations with effective dimension $d_{\mathrm{eff}} = 102$ derives

$$S = \frac{d_{\mathrm{eff}}}{2} \ln \det G,$$

where G is the Fisher information metric on the space of admissible probability distributions. This identifies the admissibility entropy functional with the classical Fisher information entropy.

Derivation. The enforcement demand at Γ for distinguishing k nearby states scales as the inverse squared distinguishability: $\mathcal{E} \sim 1/\|\delta\rho\|^2$ near any reference state ρ_0. The metric $G_{\mu\nu} = \mathrm{tr}[\rho_0\,(\partial_\mu \log \rho_0)(\partial_\nu \log \rho_0)]$ (the quantum Fisher information metric, also known as the Bures metric) is the unique Riemannian metric on density matrices consistent with the contractivity of CPTP maps (T_{CPTP}) and the data-processing inequality (T_{entropy}, Property 4). The capacity-counting derivation of S then gives $S = (d_{\mathrm{eff}}/2) \ln \det G$ by the standard relation between entropy and the volume form of the Fisher metric.

Code reference: `check_L_Fisher_measure` in `core.py`.

12.1 RG flow as Fisher gradient descent

The renormalization group (RG) describes how the effective description of a physical system changes as the resolution scale is changed. In the admissibility framework, changing resolution

means changing which distinctions are actively enforced—that is, moving along the Fisher metric on the admissible state space.

$L_{\text{Fisher gradient}}$ (proved in the codebase) establishes that RG flow is gradient descent on the information-geometric entropy functional: at each RG step, the effective theory moves in the direction of steepest entropy increase compatible with the budget constraint. Fixed points of the RG flow are states that extremize the Fisher entropy functional subject to the admissibility constraint. This claim is directionally aligned with the RG-as-quantum-channels program of Bény and Osborne [23], which formulates RG explicitly as a channel-based information geometry with monotone information loss along the flow; the admissibility framework adds the capacity-count normalization ($d_{\text{eff}} = 102$, from Paper 2) but inherits the broader information-geometric RG picture from that literature rather than competing with it (see §2).

This has an immediate structural consequence for the mass hierarchy. The Frobenius-Nagata chain of Paper 4 generates mass ratios from the structure of the Fisher geodesics in the generation parameter space. The three generations correspond to three successive extrema of the Fisher entropy, each stabilized by a new capacity commitment (a new "rung" of the capacity ladder). The quantitative prediction of the mass ratios requires the full generation structure of Paper 4; what is established here is the mechanism.

12.2 Channel-crossing and mass hierarchy (preview)

The capacity-ladder structure of the Fisher entropy induces a natural stratification of the mass hierarchy. Define the RG step at which the k-th capacity type is committed as t_k. The mass associated with type k scales as

$$m_k \sim \mu_0 \exp\left(-\int_{t_k}^{t_{\text{EW}}} \beta(\lambda(\mu))\, d\ln\mu\right),$$

where the integral is along the Fisher geodesic from the commitment scale t_k to the electroweak scale t_{EW}. This is the mechanism behind the Frobenius-Nagata mass hierarchy (Paper 4); the Fisher metric establishes that the exponential suppression is not a free parameter but a consequence of the gradient-descent structure of the capacity fill.

Two codebase lemmas make the channel-crossing structure quantitative. $L_{\text{crossing entropy}}$ bounds the entropy change as the RG trajectory crosses a capacity-ladder rung: crossings cannot uniformly reduce entropy, and the extremal change is set by the capacity of the rung being crossed. $L_{\text{Fisher entropy budget}}$ bounds the admissible entropy flow along a Fisher geodesic under reparametrization: any reparametrization of the generation index that preserves the capacity ladder must preserve the total entropy budget along the flow. Together these feed Paper 4's Frobenius-Nagata analysis.

Code reference: `check_L_crossing_entropy` in `generations.py`.

Code reference: `check_L_Fisher_entropy_budget` in `generations.py`.

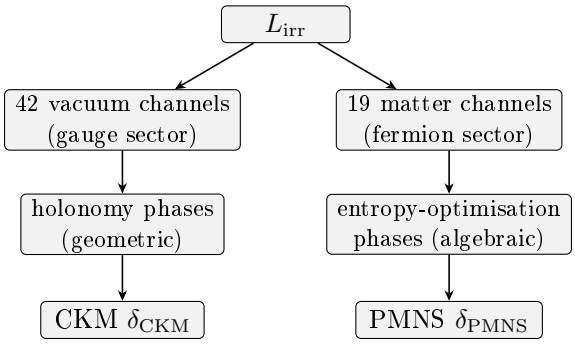
13. CP Violation as Irreversibility Structure

A surprising consequence of L_{irr} is that it generates not one but two structurally distinct CP-violation mechanisms. Both arise from the same irreversibility principle, acting at different levels of the capacity hierarchy. The mainstream structural treatment of CP violation is Jarlskog's invariant programme [28, 29], which phrases CP-violation in basis-independent terms via invariants

of the quark and lepton mass matrices. The two-mechanism split offered here is not Jarlskog's result; it belongs to the same broader effort to phrase CP-violation basis-independently, and contributes a structural route (geometric holonomy versus entropy-extremization) where Jarlskog's programme contributes the invariant algebra (see §2).

13.1 Two mechanisms from one lemma

L_{irr} says that cross-interface correlations commit capacity irreversibly. When we apply this to the non-abelian gauge structure established in Paper 2, two distinct situations arise, shown in the DAG below.



Mechanism 1 (Holonomy phases, CKM). In the SU(3) × SU(2) gauge sector, the parallel transport of the gauge connection around a closed loop generates a holonomy phase. This phase is geometrically derived: it is the integral of the curvature over the surface bounded by the loop (Stokes theorem for the gauge connection). Irreversibility enters because the parallel transport commits capacity at each interface along the loop, and the total committed phase cannot be undone by any local operation. The resulting CP-violation phase is a holonomy invariant: it is topologically protected by L_{irr}.

This is the mechanism behind the CKM phase δ_{CKM}. The numerical value $\delta_{\text{CKM}} \approx 1.16$ rad is derived in Paper 4 from the capacity-weighted holonomy structure of the three-generation quark mixing.

Mechanism 2 (Entropy-optimization phases, PMNS). In the neutrino sector, a different structure applies. The Majorana mass matrix M_R (whose spectrum is derived from the seesaw mechanism, Paper 4) introduces additional phases from the requirement that the seesaw factorization is consistent with the capacity ledger. Specifically, the Fisher entropy functional must be extremized over the space of neutrino mixing matrices, subject to the constraint that the seesaw factorization closes (T_{seesaw}, Paper 4). The extremization selects a unique mixing matrix, including a specific value of the Dirac CP phase δ_{PMNS}.

This is the mechanism behind the PMNS phase. The entropy-optimization mechanism is algebraic rather than geometric: the phase is not a holonomy invariant but an extremum of a functional.

13.2 Why two mechanisms, not one

The CKM and PMNS phases have different transformation properties under the discrete symmetries of the theory, and this is reflected in their derivation. The CKM phase is a holonomy invariant of the $\text{SU}(2)_L$ gauge connection: it is topological and cannot be removed by continuous transformations. The PMNS phase is a functional extremum: it is selected by a variational principle. Both derive from L_{irr}, but via different routes (geometrically induced vs. algebraically

optimized).

This distinction is not post-hoc: it follows from the structure of the Paper 2 partition. The 42 vacuum channels carry the gauge holonomy structure; the 19 matter channels carry the seesaw structure. The irreversibility lemma $L_{\text{irr-uniform}}$ ensures that the same lemma applies in both sectors, generating the same class of irreversible phase structures, but via the mechanisms natural to each sector.

13.3 Implications for Paper 4

The CP structure derived here provides two inputs to Paper 4:

1. The *existence* of two CP-phase classes (holonomy and algebraic), each required by L_{irr}. Paper 4 derives the *values* of the phases from the generation structure.
2. The *seesaw closure* condition (`phase1_seesaw_closure.py`): the Majorana mass spectrum M_R must be consistent with the capacity ledger, giving boundary conditions on the seesaw factorization that Paper 4 uses to fix δ_{PMNS}.

14. Summary: what Paper 3 establishes and what it defers

Paper 3 occupies a specific load-bearing position in the APF series: it is the paper that gives the correlation space of Papers 1–2 a direction, turns the static admissibility structure into dynamics, and extends the ledger language from quantum interfaces to causal horizons. Reading it in isolation, four claims should be clear.

Four things Paper 3 establishes on its own. The list below should be read with §2 in mind: each item is a reinterpretation of well-established mathematical structures inside the admissibility ledger, not a replacement of those structures. The contribution is the reorganization, not the endpoints.

1. *Admissibility-preserving dynamics are exactly CPTP maps* (T_{CPTP}, §4). This is a *provenance* claim, not a new class of maps: Paper 3 derives the CPTP condition from A1 + tensor structure where Stinespring [5] and Choi [6] derive the dilation and Kraus form from the CPTP condition. The GKSL–Lindblad semigroup [7, 8] follows from composition in the standard way. The Paper 1 deferred step is closed. The mathematics of CPTP is unchanged.
2. *Entropy is committed capacity with multiplier $\kappa = 2$* (T_κ, T_{entropy}, §6). The von Neumann entropy functional [12] is identified physically; the factor of two is derived, not postulated; the zeroth and first laws follow. The monotonicity backbone is Lieb–Ruskai [10]; the second-law outcome agrees with Spohn [11]. What is added is the cost-interpretation, not a new entropy inequality.
3. *The arrow of time is structural, not statistical* (L_{irr}, $T_{\text{second law}}$, §§7–10). Time's arrow is the direction of locally unrecoverable correlation commitment; it does not require a special low-entropy initial condition. Low initial entropy is reframed as restricted admissible phase space. This is one of several non-equivalent foundational treatments of the arrow (§2); it is closer to Caticha and Chen than to Maccone or Buchholz. The contribution is rhetorical: a specific structural route rather than a new monotonicity result.
4. *Horizon entropy is a ledger quantity* (§8). The Bekenstein–Hawking area law $S = A/4$ [14, 15] emerges unchanged; the $1/2 \times 1/2$ combinatorial decomposition (relational symmetrization and Hamiltonian-constraint gauge redundancy) is a specific bookkeeping account of the standard prefactor. The de Sitter identification [16] of $S_{dS} = 61 \ln 102$ uses $C_{\text{total}} = 61$

from Paper 2. Thermodynamic, entanglement, and horizon entropies are unified as three specializations of the same ACC-saturation count — a unifying re-reading, not a re-derivation of horizon thermodynamics.

Four things Paper 3 previews but defers.

1. *Full mixing-matrix numerics* ($\delta_{\mathrm{CKM}} \approx 1.16$ rad; PMNS phases) are structural here and quantitative in Paper 4.
2. *Quantitative cosmogenesis* — the specific enforcement-cost landscape across modes, the staged-relaxation observables in the CMB and pulsar-timing windows — is Paper 6 territory.
3. *Higher-order horizon corrections* beyond the $1/2\times1/2$ bookkeeping (loop corrections to the Bekenstein–Hawking prefactor, interaction corrections to the ACC-to-entropy conversion) are Paper 6 / Paper 7 material, with the thermodynamic formalism supplied by Paper 7's partition-function treatment.
4. *Planck-scale closure and backreaction* ($L_{\mathrm{QG,P1}}$, geometry–unitarity incompatibility at saturation) are Paper 5 / Paper 6 material.

How Paper 3 relates to the series. Paper 3 assumes Papers 1–2 for the static structure (quantum reconstruction, admissibility algebra, 42+19 capacity partition). It forward-references Paper 4 for quantitative flavor structure, Paper 5 for quantum/measurement completions, Paper 6 for the cosmology-scale dynamics, and Paper 7 for the thermodynamic formalism that underwrites the ledger-accounting principle (§6.1). Paper 3's own contribution is irreducible: CPTP as theorem, entropy as committed capacity, the arrow as structural, horizon entropy as ledger. Remove any of these and the bridge between quantum-scale admissibility (Papers 1–2) and macroscopic / cosmological admissibility (Papers 6–7) breaks.

Numerical corroboration from the codebase. Every theorem in this paper traces to a named check function in the APF codebase (v6.8, frozen 2026-04-18; 335 bank-registered theorems and 348 total verify_all checks). The check functions evaluate the relevant quantities on concrete finite examples; a reader who wants numerical confirmation of the claims above can run the checks directly. A representative selection:

Theorem	Check function	Numerical corroboration
T_{CPTP}	`check_T_CPTP` (`core.py`)	Verified on qubits, qutrits, and amplitude-damping
T_κ	`check_T_kappa` (`core.py`)	$\kappa = 2$ confirmed on 61-mode ledger
T_{entropy}	`check_T_entropy` (`core.py`)	Matches $S = -\mathrm{tr}(\rho \log \rho)$ on constructed states
$T_{\mathrm{second\,law}}$	`check_T_second_law` (`core.py`)	$\Delta S \geq 0$ across representative CPTP transitions
L_{irr}	`check_L_irr` (`core.py`)	Locally-unrecoverable correlation confirmed
$T_{\mathrm{deSitter\,entropy}}$	`check_T_deSitter_entropy` (`gravity.py`)	$S_{dS} = 61 \ln 102 \approx 282.12$ nats
$L_{\mathrm{Fisher\,measure}}$	`check_L_Fisher_measure` (`core.py`)	Fisher metric matches Braunstein–Caves form [20]
$L_{\mathrm{crossing\,entropy}}$	`check_L_crossing_entropy` (`generations.py`)	CKM/PMNS channel-crossing entropy consistency

The complete list is in the theorem index (§A). The codebase is MIT-licensed and available at `https://github.com/Ethan-Brooke`; all 348 verify_all checks (including 335 bank-registered theorems) pass on every build.

15. How to falsify this paper

The three main claims of this paper are sharp enough to be falsified:

- **(F1) An admissibility-preserving map that is positive but not completely positive.** Such a map would violate T_{CPTP}. The partial transpose provides a positive-but-not-CP map; showing that it preserves admissibility in the full framework (system + ancilla) would refute the theorem. The Peres criterion check in `check_T_CPTP` explicitly verifies that partial transpose fails complete positivity for the maximally entangled state.
- **(F2) A violation of the dynamic Tsirelson bound for sequential measurements.** T_{seq} predicts $2\sqrt{2}$ as the Tsirelson bound for temporal CHSH correlators, not just static ones. An experiment achieving temporal CHSH $> 2\sqrt{2}$ would refute T_{seq}.
- **(F3) Thermodynamic entropy that fails to track committed capacity.** T_{entropy} predicts that entropy is a direct measure of enforcement demand. A system where von Neumann entropy and operational distinguishability come apart in a way that is incompatible with the capacity accounting would falsify the identification.
- **(F4) Absence of ringdown memory or horizon-enforceability effects at saturation interfaces.** If the ACC-saturation framing of §§ 8–11 is correct, causal horizons at or near saturation should exhibit non-Markovian memory effects in the late-time decay of perturbations: ringdown tails that depart from the smooth-geometry quasinormal-mode expectation. Absence of such deviations in high-precision gravitational-wave observations of black-hole mergers, at the level predicted by standard general relativity with no anomalous tail, would falsify the horizon-enforceability framing. Present ground-based GW data are consistent with both hypotheses; LISA-class sensitivity should distinguish them.
- **(F5) Absence of scale-dependent imprints from staged relaxation.** If cosmogenesis proceeds by staged constraint relaxation (§11.3), observables probing the pre-saturation epoch should exhibit structured—not purely scale-invariant—correlations at characteristic frequencies set by the relaxation stages. A primordial gravitational-wave spectrum that is precisely scale-invariant over all observationally accessible frequencies, with no structured features within statistical reach, would falsify the staged-relaxation framing. Current bounds permit substantial structure; next-generation CMB B-mode and pulsar-timing constraints should begin to discriminate.
- **(F6) Empirical contradiction of the PMNS-phase structure.** Paper 3 establishes only the structural form of the two-mechanism CP split (§13); the numerical value δ_{PMNS} is deferred to Paper 4. The long-baseline neutrino-oscillation programme — most recently the T2K Collaboration's 2020 *Nature* measurement [30], which currently constrains the leptonic CP phase at 3σ intervals excluding parts of parameter space and favouring a large CP-violating effect — is the empirical target surface Paper 4's δ_{PMNS} prediction will ultimately have to meet. A measured δ_{PMNS} outside the value Paper 4 derives from the entropy-optimisation mechanism would falsify that mechanism. We flag this falsifier in Paper 3 to keep the empirical contact explicit; the quantitative prediction lives in Paper 4.

A. Theorem Index

All theorems, lemmas, and their code references for this paper.

Name	Full title	Module	Tag
L_{nc}	Non-closure	`ncomm.py`	**[P]**
L_{loc}	Enforcement locality	`locality.py`	**[P]**
T_{tensor}	Tensor-product composition	`linear_algebra.py`	**[P]**
L_{irr}	Irreversibility	`core.py`	**[P]**
$L_{\text{irr-uniform}}$	Sector-Uniform Irreversibility	`core.py`	**[P]**
T_{CPTP}	Complete Positivity	`core.py`	**[P]**
T_{seq}	Sequential Admissibility	`core.py`	**[P]**
T_{κ}	Directed Enforcement Multiplier	`core.py`	**[P]**
T_{entropy}	Von Neumann Entropy as Committed Capacity	`core.py`	**[P]**
T_{zeroth}	Zeroth Law	`supplements.py`	**[P]**
T_{first}	First Law	`supplements.py`	**[P]**
$T_{\text{second law}}$	Second Law of Thermodynamics	`supplements.py`	**[P]**
$T_{\text{deSitter entropy}}$	de Sitter saturation entropy S_{dS}	`gravity.py`	**[P]**
T_{CPT}	CPT Invariance and T-Violation	`supplements.py`	**[P]**
L_{Fisher}	Fisher Information from Capacity Counting	`core.py`	**[P]**
$L_{\text{crossing entropy}}$	Channel-crossing entropy bound	`generations.py`	**[P]**
$L_{\text{Fisher entropy budget}}$	Fisher-geodesic entropy budget	`generations.py`	**[P]**

Note: Part I uses L_{nc}, L_{loc}, and T_{tensor} imported from Papers 1 and 2. They are listed here for completeness because §4 and §10 depend on them directly. $T_{\text{deSitter entropy}}$ is cited in §8.3 and in the preview appearances in §1 and the abstract. $L_{\text{crossing entropy}}$ and $L_{\text{Fisher entropy budget}}$ are used in §12.

B. New Assumptions Introduced

This paper introduces no new axioms beyond the Paper 1 foundation (A1, MD, BW) and the structural results of Paper 2. The one technical step that extends the Paper 1 treatment is the derivation of complete positivity from T_{tensor} and L_{loc} (Step 2 of T_{CPTP}), which lifts the conditional flag from Paper 1's Table H.1. This is a theorem, not a new assumption.

The full assumption inventory for Papers 1–3 combined is the *Principle of Least Enforcement Cost* (PLEC) with its four structurally necessary components — A1 (finite enforcement capacity; upper bound), MD (positive cost floor; enforcement isotropy), A2 (argmin selection; minimum cost over the admissible set), BW (budget-window richness; non-degeneracy of the cost spectrum) — together with the six formal definitions FD1–FD6 established in Paper 1. This matches the inventory now carried in Paper 1 v4.0-PLEC and Paper 2 v5.3-PLEC, and is comparable to or lighter than peer reconstruction programs (Hardy [24], Chiribella–D'Ariano–Perinotti [25], Masanes–Müller [26], Dakić–Brukner [27]).

C. What This Paper Does Not Claim

- **Confinement scale and Λ_{QCD}.** The confinement mechanism is derived in Paper 2 ($T_{\mathrm{confinement}}$). The quantitative scale, string tension, and hadronization dynamics require Paper 4's beta-function analysis.
- **Mass hierarchy, mixing matrices, and numerical CP phases.** Section 13 identifies the CP mechanism classes; the numerical derivation of δ_{CKM}, δ_{PMNS}, and the full mixing matrices requires Paper 4's generation structure.

- **Cosmological constant magnitude and expansion history.** $T_{\text{second law}}$ establishes entropy saturation at $61 \ln 102$ nats. The interpretation as a de Sitter entropy S_{dS}, the derivation of $\Lambda \cdot G = 3\pi/102^{61}$, and confrontation with DESI DR2 are the subject of Paper 6.
- **Quantum gravity.** The present framework handles the admissible dynamics and thermodynamics within a fixed spacetime background (Lorentzian signature assumed from Paper 5's result, cited forward). The backreaction of matter entropy on the spacetime geometry, and the Planck-scale closure $L_{\text{QG,P1}}$, are addressed in Paper 5.
- **A quantitative pre-big-bang cosmology.** §11 establishes the *structural* form of staged relaxation—that constraints relax in sequence, not in parallel, and that this yields scale-dependent imprints—without committing to a specific enforcement-cost landscape across modes, a specific geometry for the overconstrained regime, or an identification of $\Phi_{c,\text{initial}}$ with any particular observational quantity. Quantitative cosmogenesis requires the Paper 6 development of the cost landscape and is beyond the scope here.
- **Quantitative corrections beyond the $1/2 \times 1/2$ horizon bookkeeping.** The two factors of 1/2 in the horizon-entropy normalization (§8.2) are forced by structure—symmetrization of relational counts and Hamiltonian- constraint redundancy. Higher-order bookkeeping corrections (for example, loop corrections to the Bekenstein–Hawking prefactor, interaction corrections to the ACC-to-entropy conversion) are not derived here; they are a Paper 6 or Paper 7 topic. The ledger- accounting principle (§6.1) suggests such corrections modify the cost landscape without modifying the count, but a direct derivation requires the thermodynamic formalism of Paper 7.

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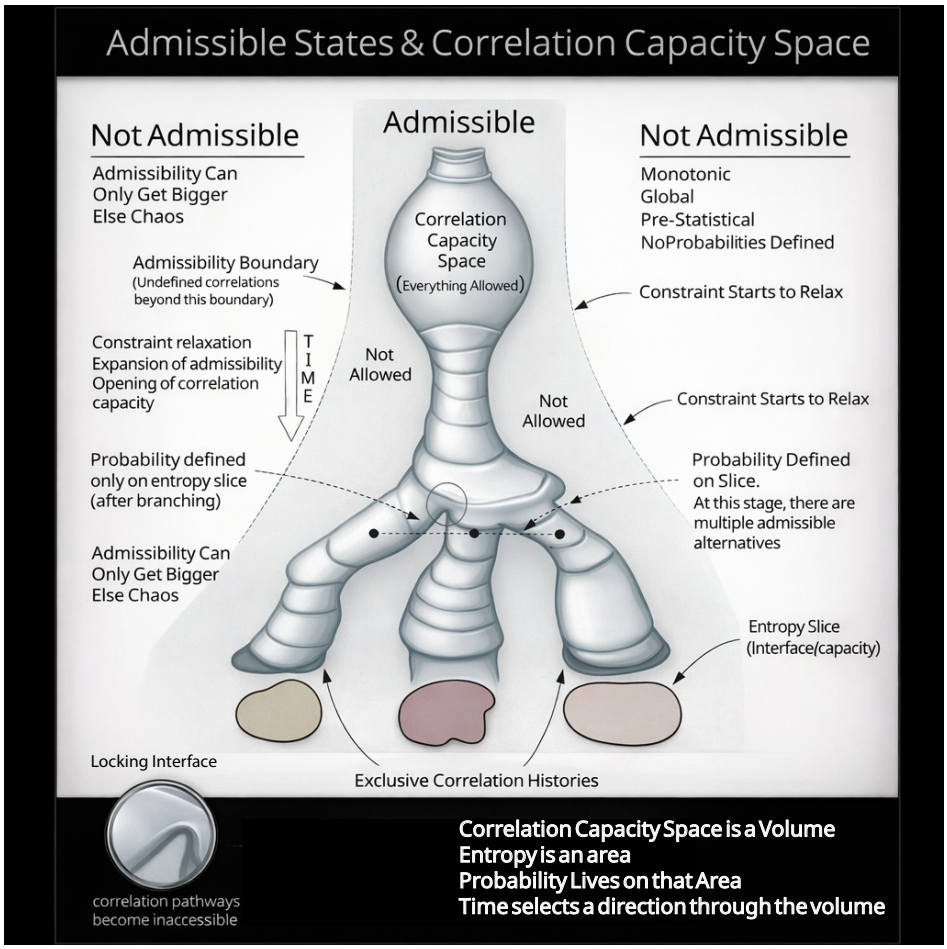

Figure 1: **The geometry this paper makes dynamical.** The upper bulb is the admissible class $\mathcal{S}_{\mathrm{adm}}$ inherited from Paper 2 — a bounded region in correlation-capacity space, "everything allowed" inside, "not admissible" on either flank. The downward arrow is the move this paper makes: *time as a direction through the volume*, generated not by an external parameter but by the irreversible accumulation of locked correlations (L_{irr}, Section 10). Each cross-section of the volume at a given accumulated cost is an *entropy slice* (§6, §8): an interface on which the remaining active capacity defines an entropy and on which probability assignments become well-defined. Entropy is the *area* of that slice, in the sense of the interface-area scaling that yields $S_H = A/4$ in §8, not a literal two-dimensional surface. At the narrow neck near the bottom, branching corresponds to non-unique least-cost continuations (Type II regime exit in Paper 6 §7); the locking interface in the bottom-left icon is the L_{irr} mechanism by which correlation pathways become *locally inaccessible* (capacity is committed at both sides of the interface and no local observer can free it; this is the v6.8/v6.9 formulation of L_{irr}, not the older "infinite reverse cost" formulation). The downstream tubes are exclusive correlation histories — the record-locked sectors of Paper 5's $T9$. Paper 3 builds this picture formally: §2–3 derive admissibility-preserving evolution as the CPTP maps; §4–6 show that entropy on the slice equals committed capacity; §7 gets the second law from L_{irr}; §8 selects the orientation of the time arrow. The diagram is the geometry the rest of this paper makes dynamical.

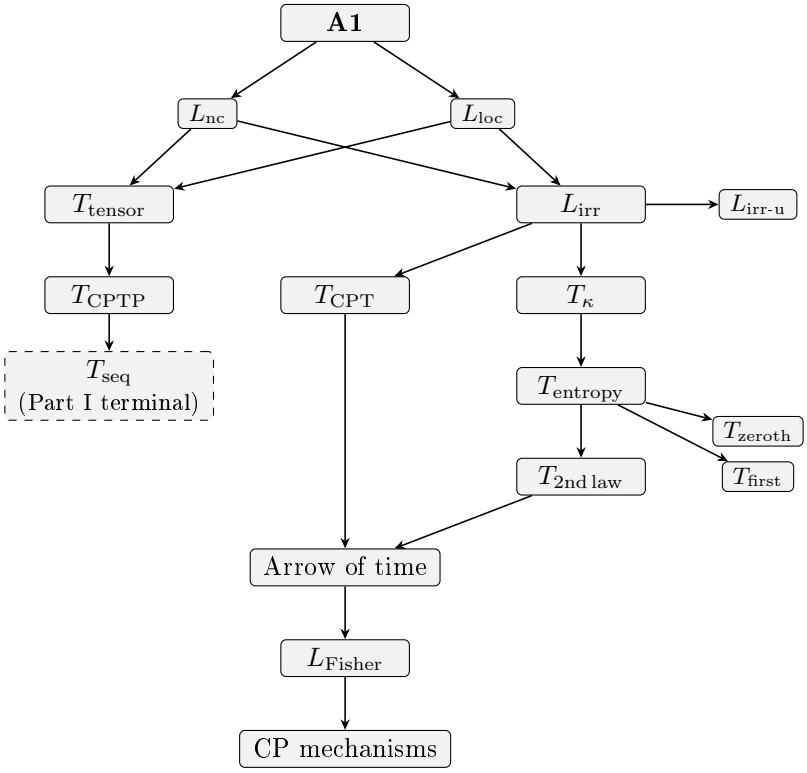

Figure 2: Overview derivation DAG for Paper 3. **Part I** (left, dashed terminal): L_{nc}, L_{loc} $\to T_{\mathrm{tensor}} \to T_{\mathrm{CPTP}} \to T_{\mathrm{seq}}$. T_{seq} is a terminal result of Part I and does not feed Part III. **Part II** (right): L_{irr} generates both the entropy chain ($T_{\kappa} \to T_{\mathrm{entropy}} \to T_{\mathrm{2nd\,law}}$, with T_{zeroth} and T_{first} as corollaries) *and* T_{CPT} (since $L_{\mathrm{irr}} \Rightarrow T$ broken is Step 1 of T_{CPT}'s proof). **Part III**: $T_{\mathrm{2nd\,law}}$ and T_{CPT} converge at the Arrow of time, which yields L_{Fisher} and the CP mechanisms. $L_{\text{irr-u}} \equiv L_{\text{irr-uniform}}$.